

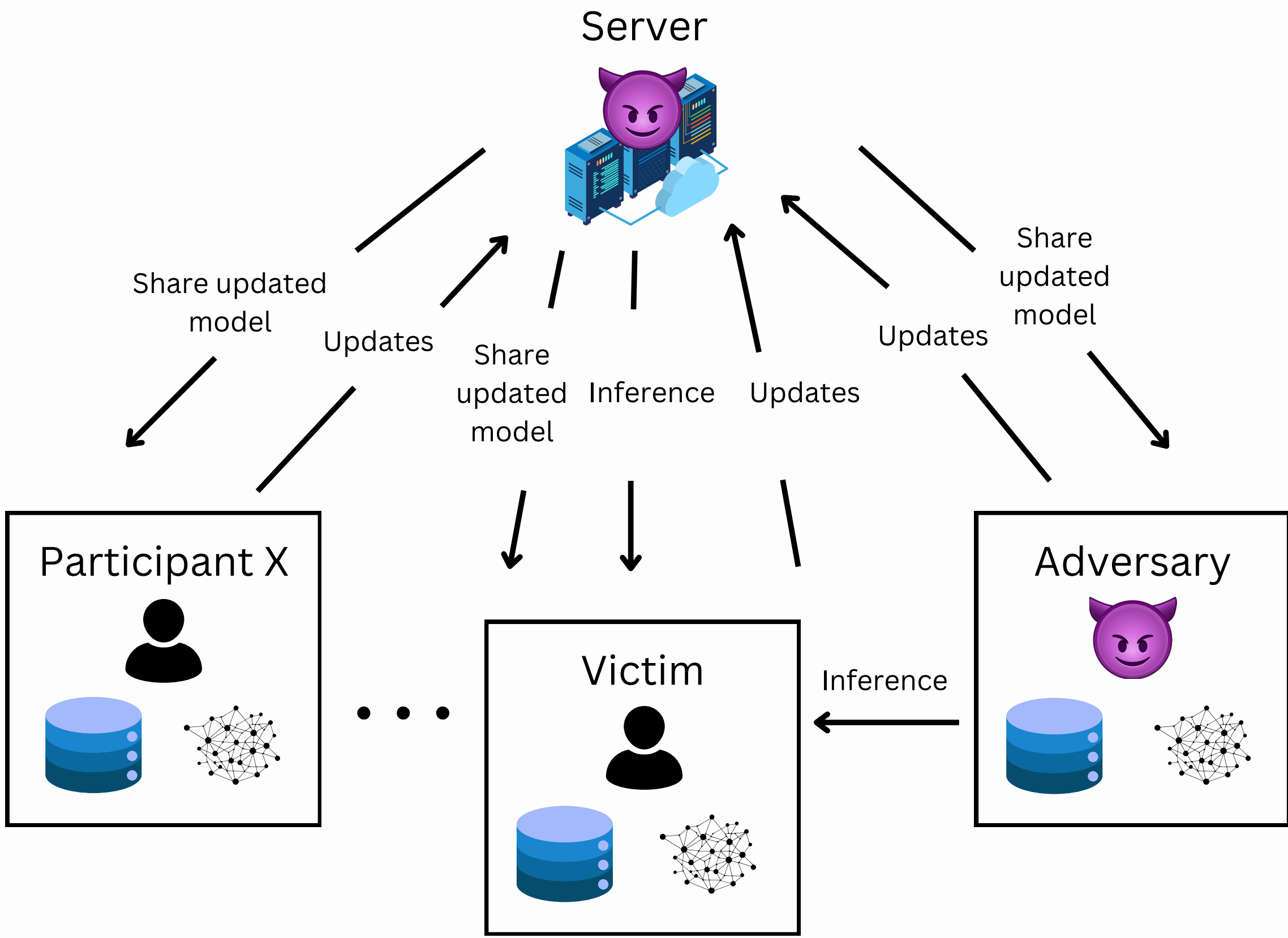
Defending Federated Learning Against Inference Attacks: Techniques and Approaches

Mohammad Kiyani, Dev Mehta, Amanda Sherman



Introduction

- Federated Learning (FL) is a decentralized machine learning approach where multiple clients collaboratively train a model without sharing local data
- Each participating device or node trains a local model on its data
- Devices send only the model updates (e.g., weights and gradients) to a central server
- The central server aggregates these updates to improve the shared global model [1]
- This process is iterated multiple times
- Efficient communication protocols are used to reduce the amount of data transferred during model updates
- Unlike centralized ML models, the central server does not have access to individual users' data and users cannot access training data from other users
- FL is widely used in smart healthcare and financial forecasting



Contributions:

- Analyze Membership Inference Attacks in the context of Federated Learning
- Examine strengths and weaknesses of commonly used datasets used to test the effectiveness of MIAs against FL models
- Assess mitigation strategies to prevent MIAs against FL models

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Membership Inference Attacks

- Membership Inference Attacks (MIAs) occur when an adversary can determine whether a specific data point was part of the training dataset used to train a machine learning model.
- Inference attacker first builds and trains collection of shadow models w/ behaviors comparable to target model, then uses shadow models to generate attack dataset for training attack model
- Trained attack model can infer whether sample belongs to training set of target model data
- After FL model is trained, global model is produced which is made up of training outcomes of all participants' data that is sourced from multiple rounds of aggregation
- Attack models are trained on shadow model outcomes and thus can effectively infer which target samples are “in” and “out” of global model's dataset [3]
- Existing Membership Inference Attacks (MIAs) can perform poorly because of a lack of attack data stemming from the fact that each participant is independent in FL [4]
- To overcome this limitation, attackers can use a Generative Adversarial Network (GAN) to enrich attack data and increase data diversity, thereby boosting the model's performance [2]
- Active Learning can be used to complement GAN in order to better label newly generated data in an approach called Active Learning Generative Adversarial Network (ALGAN) [5]
- Two different types of MIAs are passive and active attacks
- Passive attacks focus on getting membership leaks based on information that can be accessed without compromising the normal training process
- Active attacks modify FL updates and therefore increase the trained model's vulnerability to attacks

References

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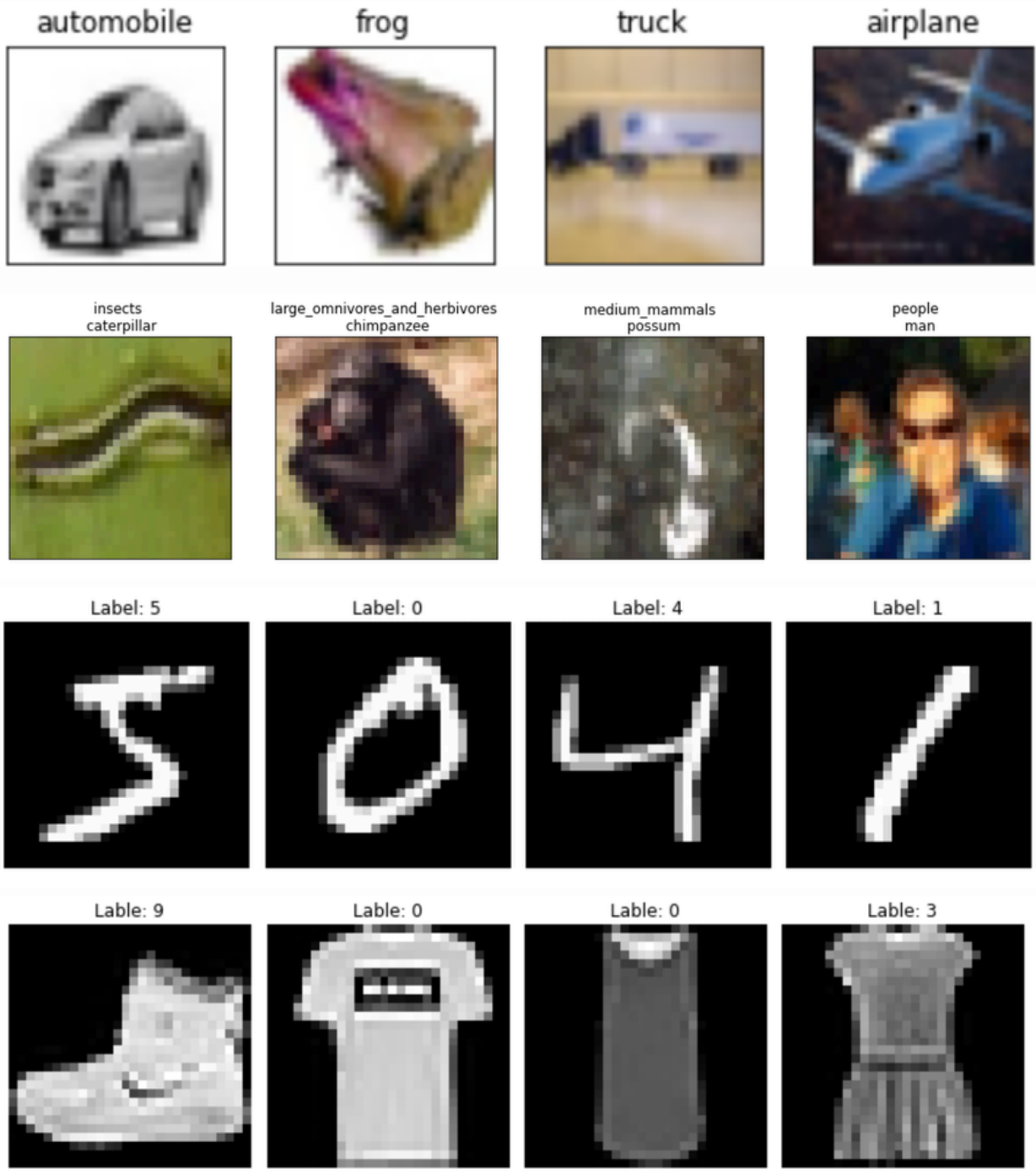
Data Analysis

Datasets Analyzed:

- CIFAR-10
- CIFAR-100
- Purchase-100
- MNIST
- FMNIST
- Dermnet
- Texas-100
- Adult Dataset

Alternative Datasets:

- CIFAR-AUG



Models with more classes, such as CIFAR-100 compared to CIFAR-10, tend to reveal more information, leading to higher membership inference attack accuracy. This is because models with more classes must remember more details to classify inputs accurately, making them more susceptible to membership inference attacks. For instance, CIFAR-100, which is more overfitted and has more classes, shows higher attack accuracy than CIFAR-10, even with fewer training records per class.

Mitigation

- Differential Privacy (DP) provides a provable defense against MIAs by adding noise to the training process [6] **Definition 1**
- DP struggles to balance privacy-utility tradeoff
- Differentially Private Stochastic Gradient Descent (DP-SGD) involves adding Gaussian noise to gradients during model training which balances security and model accuracy [6]
- Secure Multiparty Computation (SMC) involves multiple distributed participants jointly computing a common function, where each participant can only obtain its own information from the output, and the result of the protocol can be delivered securely
- Homomorphic encryption is an encryption scheme that allows complex mathematical operations to be performed on cipher-text without changing the nature of the operation, defending against MIAs in FL with minimal effect on accuracy [6]
- Client-level DP hides the membership of individual data by clipping the gradients at the client level and adding Gaussian noise [7]

Definition 1: (Differential Privacy). A randomized algorithm $\mathcal{M}$ with domain $\mathbb{N}^{|\mathcal{X}|}$ is (ϵ, δ) -differentially private if for all $\mathcal{S} \subseteq \text{Range}(\mathcal{M})$ and for all $x, y \in \mathbb{N}^{|\mathcal{X}|}$ such that $\|x - y\|_1 \leq 1$:

$$\Pr[\mathcal{M}(x) \in \mathcal{S}] \leq \exp(\epsilon) \Pr[\mathcal{M}(y) \in \mathcal{S}] + \delta,$$